ParkSmart - Member 4 Presentation Script

Smart Parking, Traffic Enforcement & Citizen Reporting System (PS-03)

Member	Member 4	**Role**	Operations & Tech Lead
Name	Susmitha Sree M	**Slides**	Slides 9 to 12
Your part	Command dashboard, architecture and technical design	**Speaking time**	about 1 minute 45 seconds

1. What to say

Speak slowly, look at the judges, and use your own words if you forget a line. The last sentence of your part hands over to the next member.

Slide 9 - Dashboard

"This is the Traffic Police command dashboard. It shows free slots, full lots, open reports, challans, fines and response time. Each zone gets a congestion index from parking occupancy and recent violations. Most importantly, it recommends actions: for example, send the tow vehicle to Mission Street between 4 and 7 pm, and guide drivers to a lot nearby that usually has space. The officer verifies a report and issues a challan with a QR code in one click, and repeat offenders are listed with unpaid fines."

Slide 10 - How it runs

"It runs on any laptop with one double-click. The phone or browser talks to a Node.js server, which runs our rules engine on a data file. There is no database server and no internet needed. Sensors and cameras can later send counts to the same API."

Slide 11 - Technical layers

"There are four layers: the web client with maps and charts, a server built only with Node.js built-in modules, the rules engine, and the data layer with an audit log. 57 automated checks test every feature. The same engine also runs inside our Android app, so the phone works with no server at all."

Slide 12 - Algorithms

"These are the methods: hourly demand forecast, street matching by distance, duplicate linking within 30 minutes, rule-based priority, the CCTV dwell-time rule, the congestion index, and the peak-window recommendations. Now Kavini will talk about security and field use."

2. Your job in the live demo

- On the laptop (logged in as traffic): open **Violation Reports**, click the citizen report Akash just sent.
- Click **Verified on spot**, then **Issue challan**, then **Print challan** to show the QR challan.
- Go to **Command Dashboard** and read one recommended action aloud. Hand over to Kavini.
- Later, if there is time: **Violation Reports → Import phone sync**, pick Kavini's sync file, and show the new report (ID starts with M).

Live demo order (about 4 minutes): Akash uses a phone to find parking and report a violation, Vaishnavi runs CCTV detection on the laptop, Susmitha Sree opens the report, issues the challan and shows the recommended actions, Kavini tracks the report and records a violation on the ParkSmart Mobile app, Siddharth closes and invites questions.

3. Questions you answer

Q1. How is the congestion index calculated?
60% is how full the parking in that zone is, and 40% is the number of violations in the last 2 hours. 75 and above is Severe.
Remember: 60 parking + 40 violation.

Q2. How do recommendations choose the time?
For each hotspot street, we slide a 3-hour window over the last 7 days of reports and pick the busiest window. Then we find the nearest lot within 900 m that is usually under 80% full at that time.
Remember: Busy 3 hours + nearby free lot.

Q3. Why Node.js and JSON instead of a database?
For the prototype, it needs zero installation and runs offline on one laptop. For the whole city, the same API can move to PostgreSQL.
Remember: Demo-ku simple; scale-ku PostgreSQL.

Q4. How do live updates reach the phone?
The server uses server-sent events. When a count changes, it pushes the new numbers to every open page, so no refresh is needed.
Remember: Server push pannum.

4. Numbers to remember

- 20+ API endpoints, 57 automated checks
- Congestion index = 0.6 occupancy + 0.4 violations
- Runs on port 5053 (CrimeMap is 5050)

5. Practice checklist

[ ] Read your part aloud 3 times with a timer.
[ ] Practise your demo clicks on the laptop or phone once.
[ ] Answer your questions without looking at this sheet.
[ ] Full team rehearsal on 12 Oct, twice.